

NLPTEL

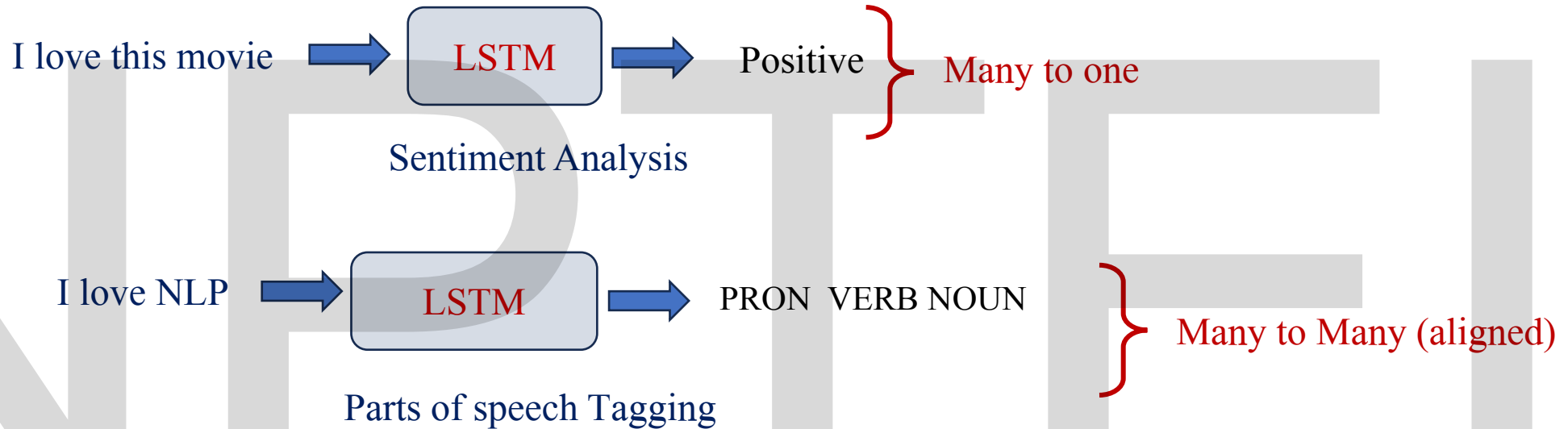
LSTM Encoder Decoder Structure

Contents

- ✓ Introduction to Encoder Decoder Architecture
- ✓ Encoder LSTM
- ✓ Decoder LSTM
- ✓ Teacher forcing method
- ✓ Training and inference of Decoder
- ✓ Limitations of Encoder Decoder architecture

Motivation for Encoder Decoder LSTM

A single LSTM is a network that processes a sequence and produces outputs within the same sequence.

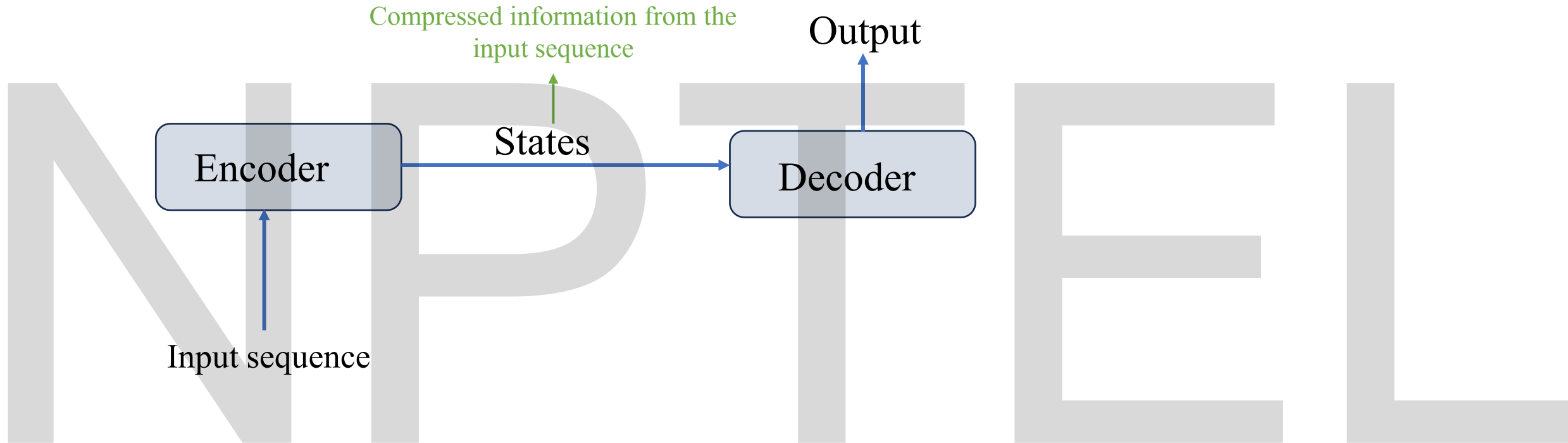


Examples:
Machine Translation (English $\rightarrow$ French)



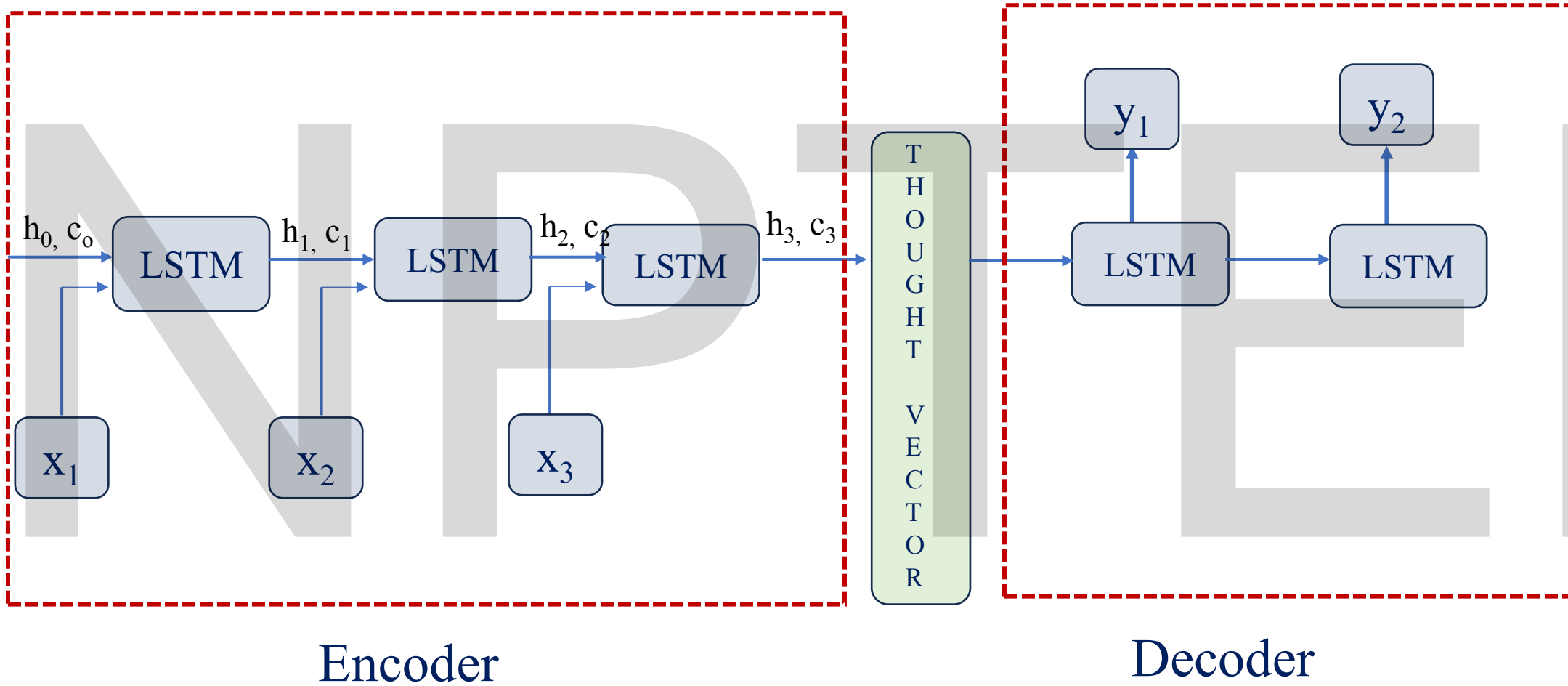
Sequence-to-Sequence Modeling (can be handled by encoder decoder LSTM)

Encoder Decoder Architecture (Seq2Seq with LSTM)

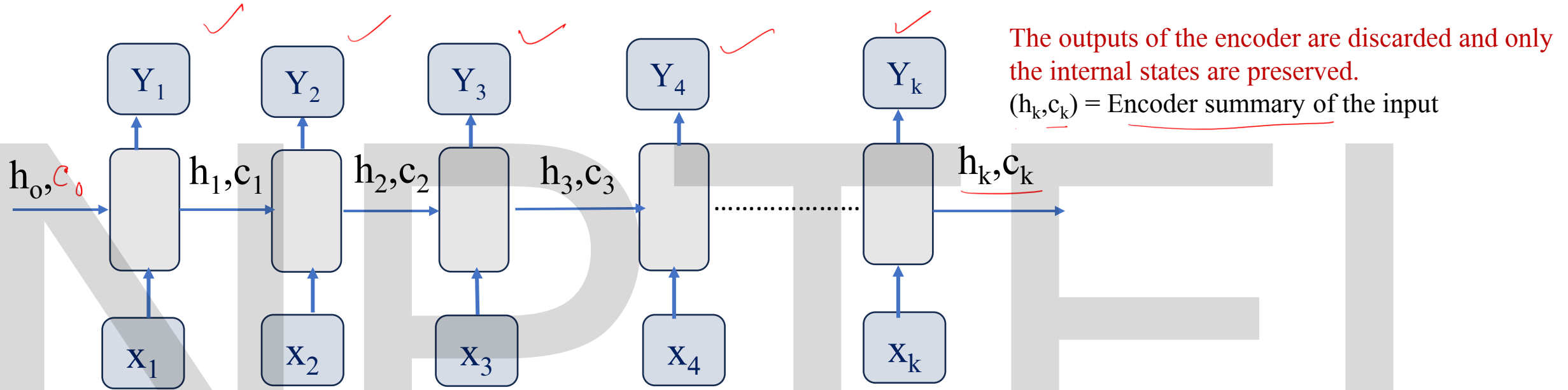


- Both Encoder and Decoder are made up of LSTM
 - States consist of
 - ✓ Hidden state
 - ✓ Cell state

Encoder Decoder Architecture



Encoder LSTM



X_i → Input sequence at time step i (word embeddings or feature vector)

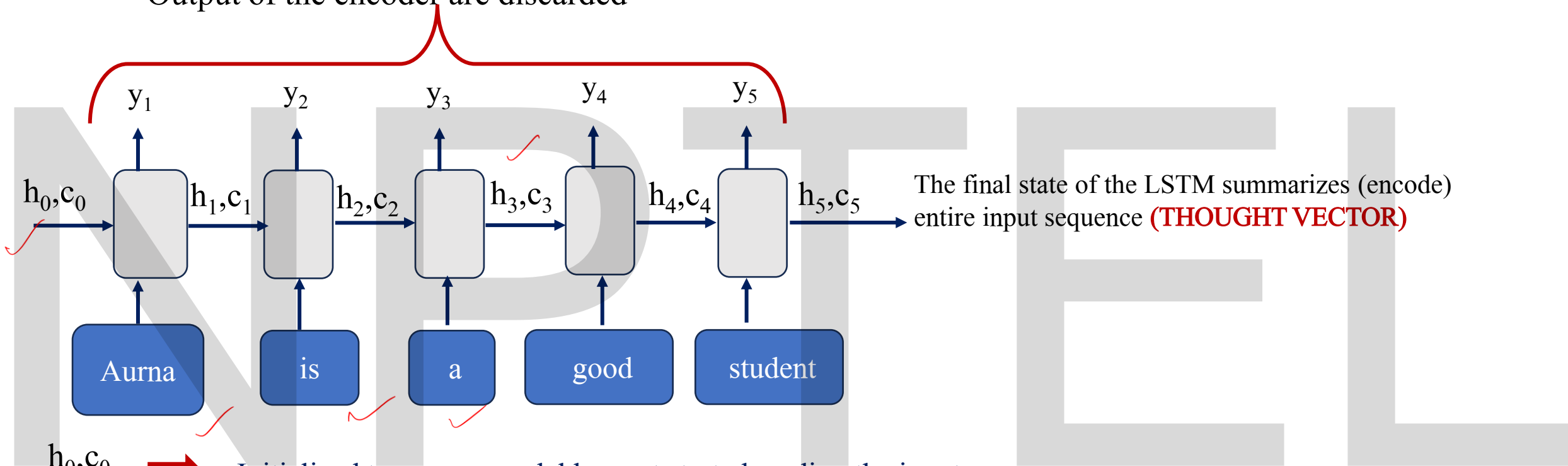
h_i → Hidden state at time step i

c_i → Cell state at time step i

Y_i → Output /predictions sequence at time step i

Encoder LSTM

Output of the encoder are discarded



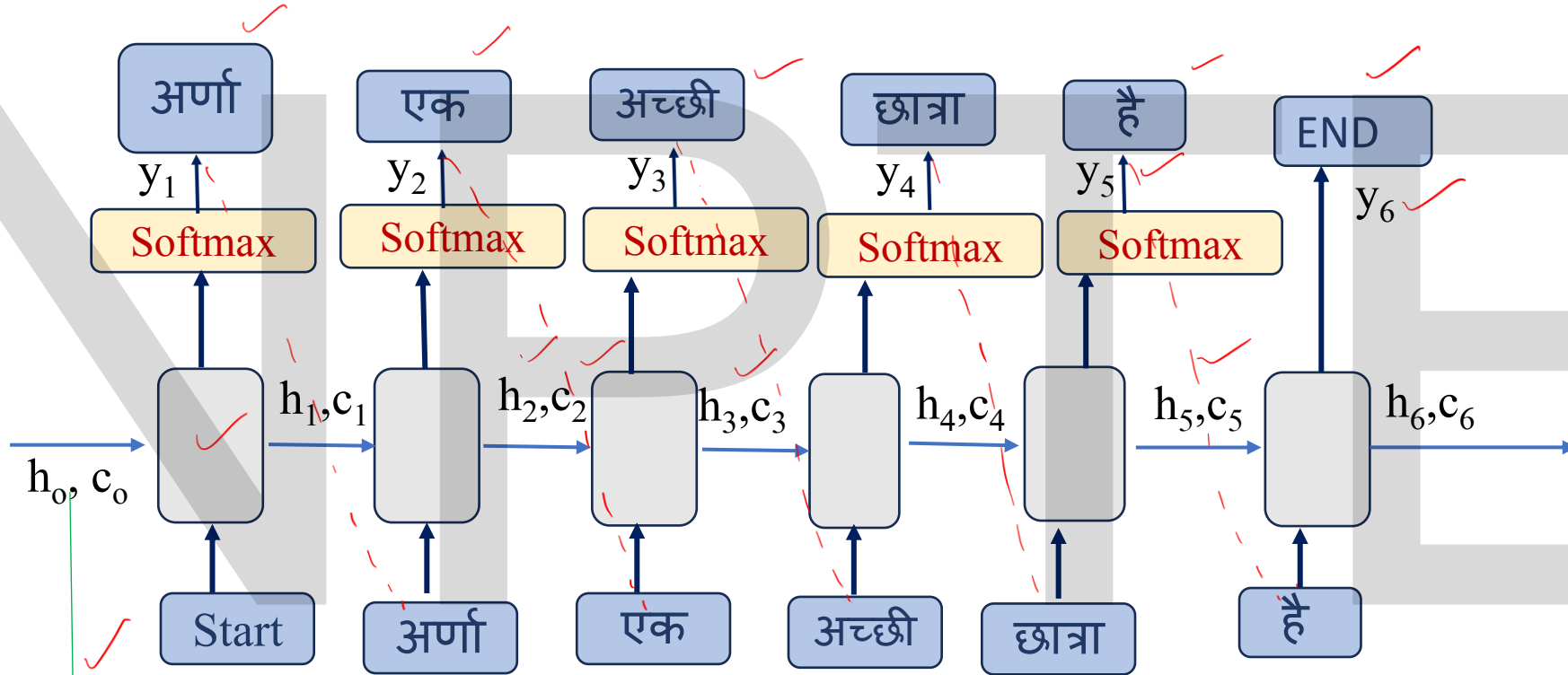
h_0, c_0 → Initialized to zero as model has not started reading the input

h_i, c_i → LSTMs maintain two states [hidden state h and cell state c] at each time step. Combined together are the internal states of LSTM at each time step i . They remember what the LSTM has read till now.

h_3, c_3 → Vectors remember that network read "Aurna is a"

Decoder LSTM

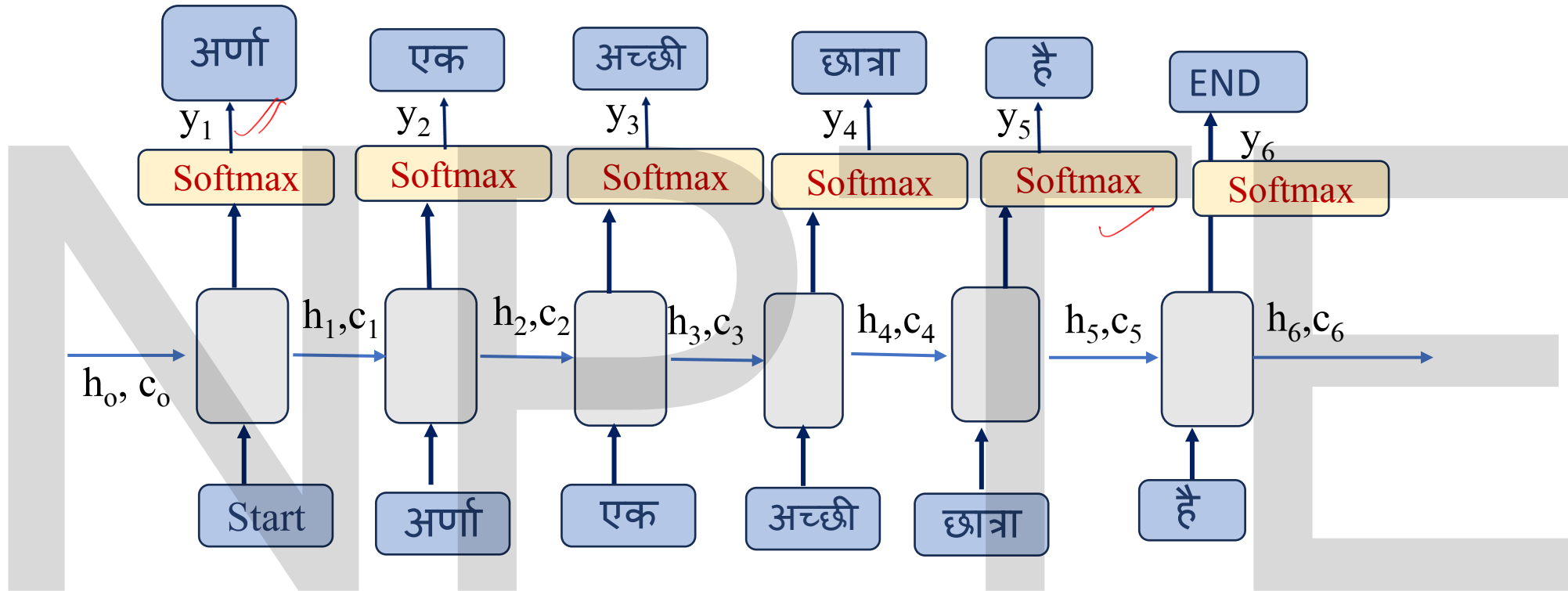
- ✓ The initial state of the decoder LSTM are initialized to the final state of the encoder LSTM.
- ✓ As the encoder scanned the input sequence word by word, similarly decoder will generate the output sequence word by word



These states are the
output states of the
encoder

Two tokens START and END are added to the output sequence

Function of Softmax in the Decoder LSTM

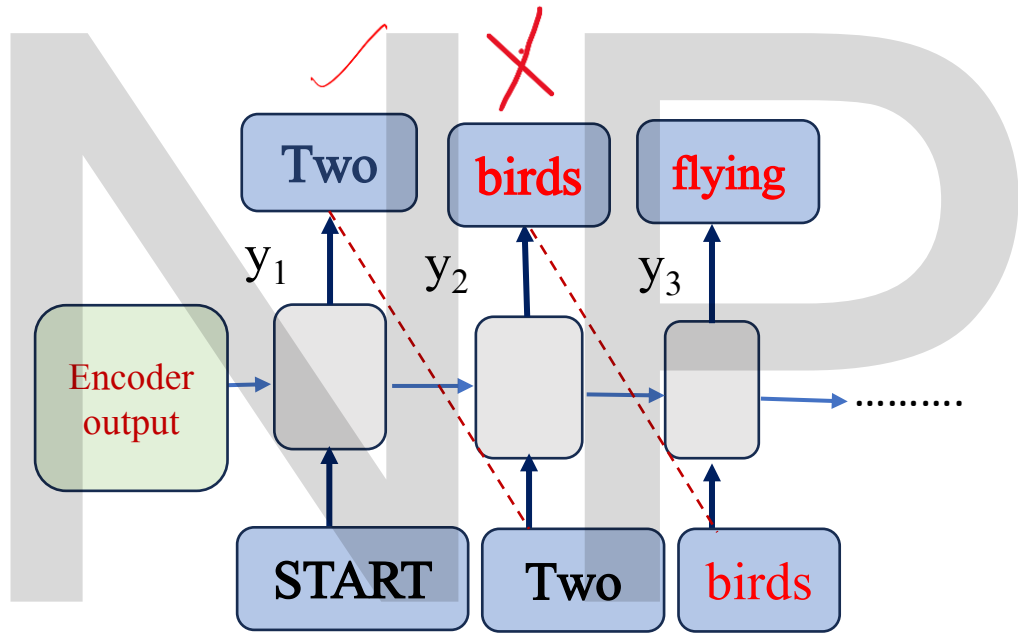


Y_i s are the predictions of Decoder LSTM model at each time step.

Each Y_i is actually a probability distribution over the entire vocabulary which is generated using a softmax activation function

Teacher Forcing

Suppose we are trying to do an image captioning task:



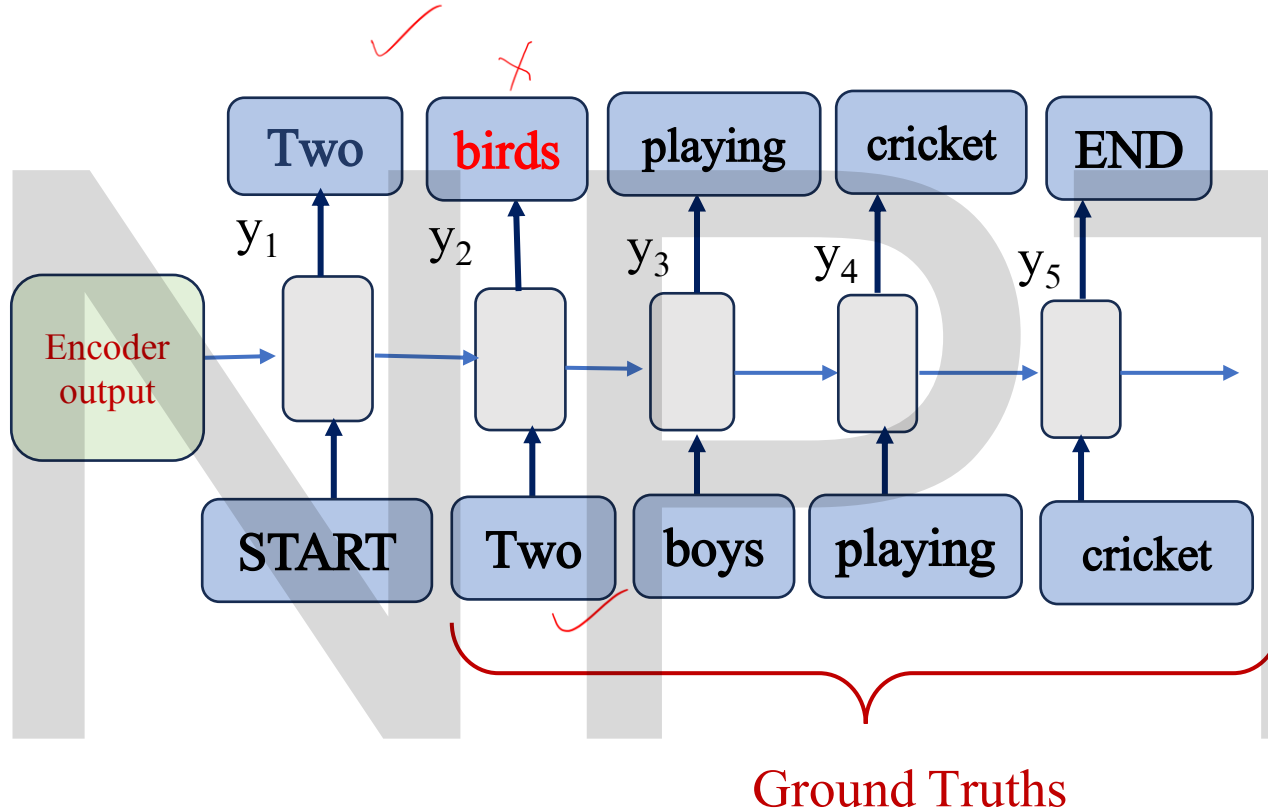
Normal Decoder Training (without Teacher forcing)

Mistake in prediction of one word affects the entire prediction of the image caption



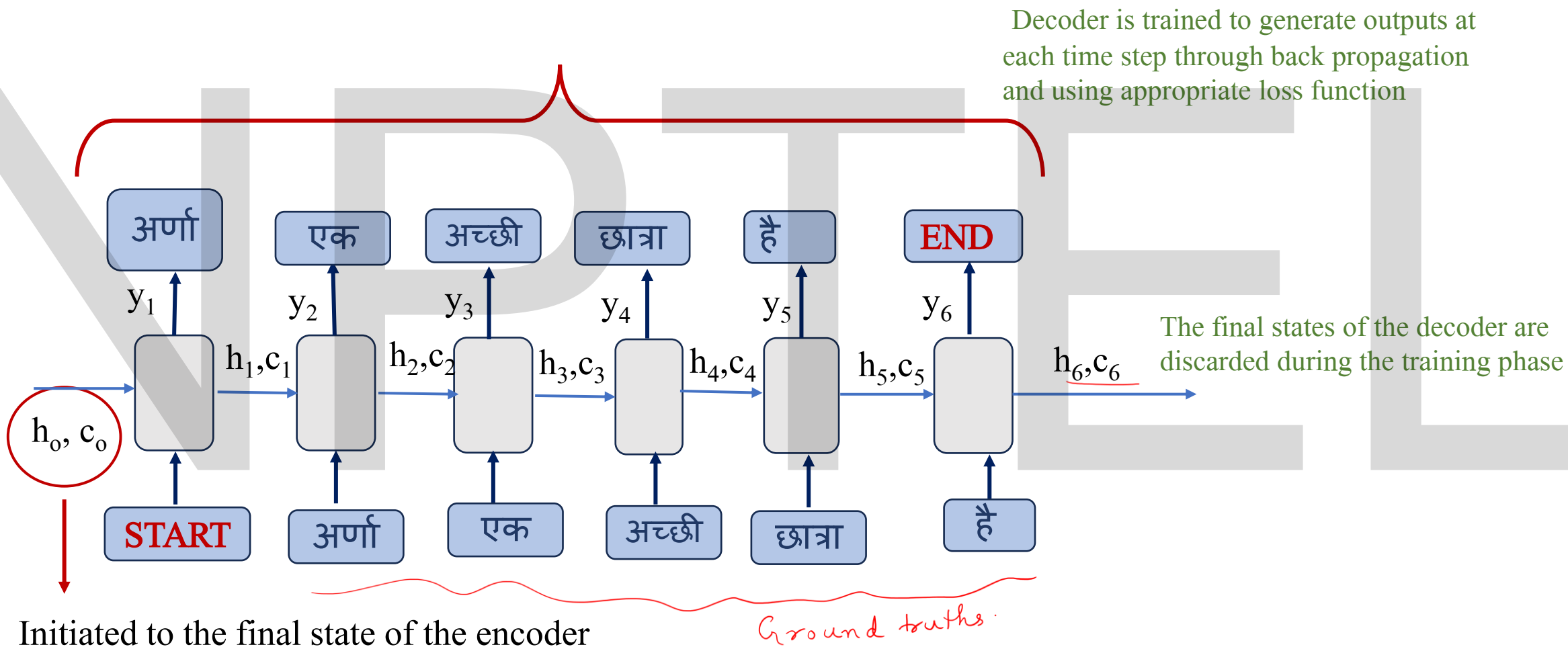
“Two boys playing cricket”

Teacher Forcing



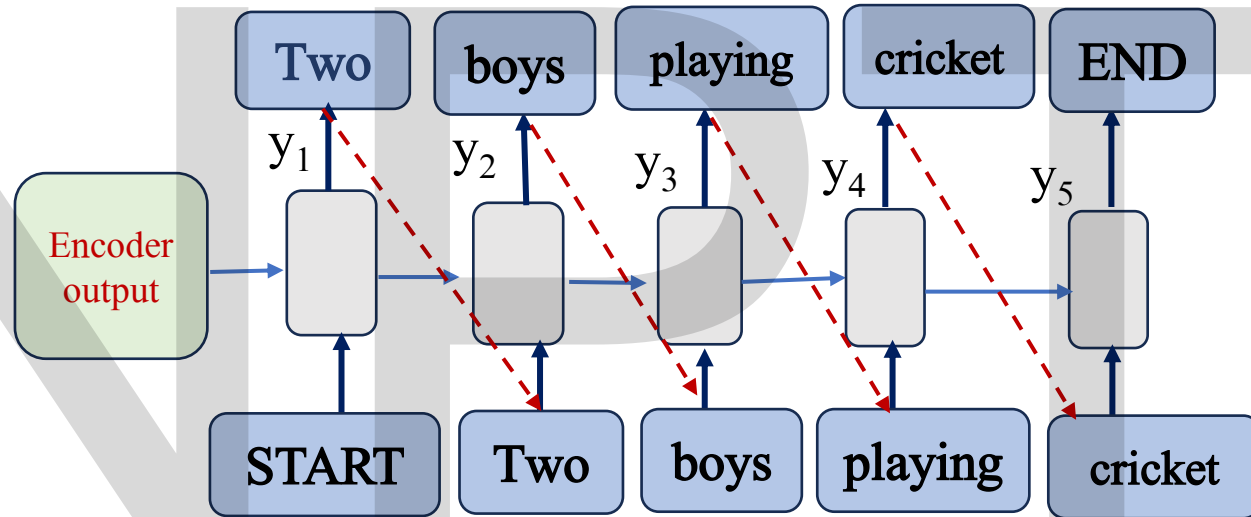
During training time, the input at each time step to the decoder are the actual output (and not the predicted output) of the previous time step – **Teacher Forcing**

Decoder LSTM

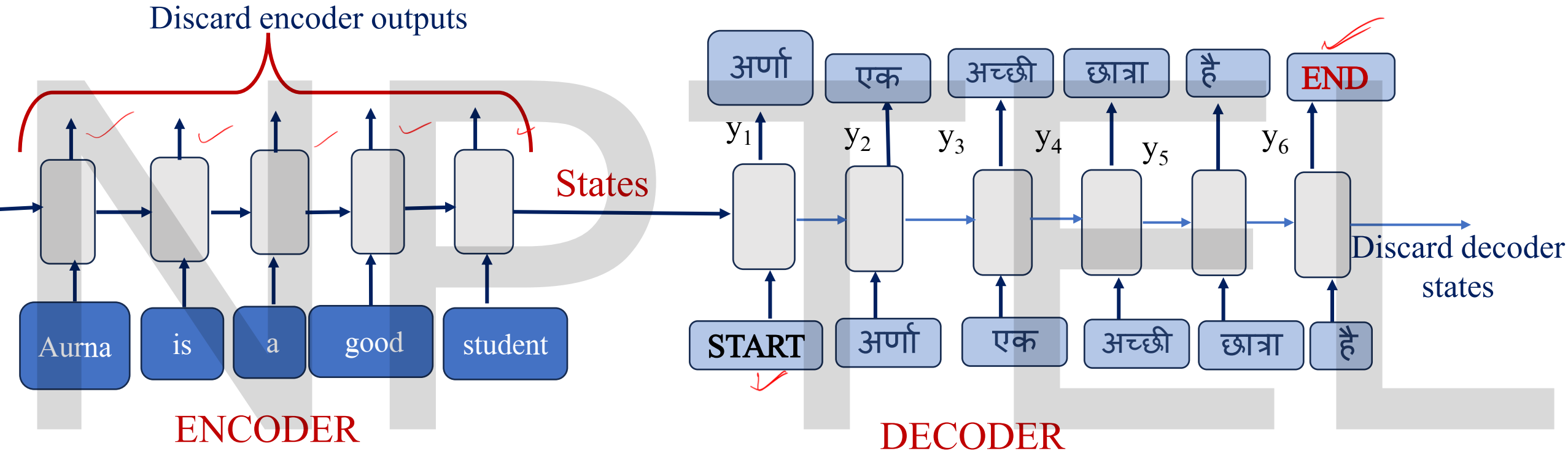


Inference from the Decoder

During inference time, there are no ground truths, hence predicted output is given as the input to the next sequence



Encoder Decoder Architecture



Limitations of Encoder Decoder Architecture

- ✓ Encoder steps needs to represent the entire input sequence $x_1, x_2 \dots x_n$ as a single vector c .
- ✓ This may cause information loss as all information needs to be compressed within c
- ✓ Decoder needs to decipher the passed information from this single vector, which is a highly complex task

Solution:

Attention Mechanism!!

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Summary

Next Session

Attention Mechanism in LSTM

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Thank you